HackInsight '26 · Karachi Air Quality · Team Haseen

Track: Air quality · Muhammad Taha Farrukh, Muhammad Bilal Rasheed, Muhammad Uzzam Siddiqui, Uzair Ahmed · github.com/UzairAhmed87/HackInsight-TeamHaseen (tag v1.0-final) · hackinsight-air-quality-pakistan.onrender.com

Problem and primary user

Karachi residents, and the school heads, clinic staff and journalists who advise them, see PM2.5 numbers with no source, no quality check and no admission of uncertainty. We built a traceable, reproducible warehouse and agents that explain it without inventing anything.

Dataset: 115,146 gold rows from three original publishers, rebuilt with `make data`

Source (publisher)	Access / licence	Rows	Coverage
PAQI Karachi daily PM2.5 (Pakistan Air Quality Initiative)	Zenodo DOI 10.5281/zenodo.15845428 · CC BY 4.0	3,009 days	2016-10-06 → 2024-12-31
U.S. Consulate General Karachi monitor (U.S. Dept. of State / EPA AirNow)	OpenAQ location 8156 · US Public Domain	46,937 hours	2019-05-22 → 2025-03-04
Urban Unit AirGradient network, 8 sensors (The Urban Unit)	OpenAQ v3 hourly · CC BY 4.0	65,200 hours	2025-06-19 → refreshed hourly

The raw files are kept verbatim and never overwritten, each with a SHA-256 provenance sidecar. Every gold row has a stable SHA-256 `row_id` and the hash of its raw file. Licence and provider are re-read from source metadata on every run. **Cross-source check:** on 1,399 shared days, PAQI vs Consulate Pearson r = 0.87, mean difference +3.43 µg/m³; 63 days are flagged for large disagreement. PAQI does not publish its monitor set, so we report agreement instead of assuming independence. No synthetic or personal data is used.

Data Quality Agent (deterministic)

- **PAQI:** 11 pass / 4 warn / 0 fail. FIX 305 (`22.0`→22, original kept). FILL not applied to 6 blank days. FLAG 12 (undocumented methodology, staleness).
- **Consulate:** 10,519 null hours, FILL not applied. One 985 µg/m³ BAM error value excluded. 8 hours > 500 kept for review. Daily mean only with ≥ 18 valid hours (EPA 75%).
- **Urban Unit:** 27,294 absent sensor-hours not imputed; checks for frozen values, low coverage, stale sensors and the low-cost-sensor caveat.
- Critical schema, duplicate, hash or licence failures stop the run. Every decision is logged with a reason and row count.

Model vs declared baseline

Next-day PAQI value, OLS on 1/2/7-day lags vs persistence; frozen chronological splits (test 2023-10-07 → 2024-12-31, n = 436).

Test	MAE	RMSE
Linear model	**9.90**	**16.33**
Persistence	10.15	16.94

Uncertainty: the validation-residual 5–95% interval (−23.8, +29.5) covers 91.5% of test days. **Errors by group:** winter 17.46 vs 18.63; previous day > 75 µg/m³ 18.86 vs 20.10; monsoon 3.56 vs 3.22 (persistence better). No significance is claimed.

Insight Agent (RAG) and generative layer

A deterministic router retrieves evidence from the gold tables and reports, and computes every number at query time. It returns cited evidence (file, locator, row IDs), limitations and `data_as_of`. It says *"I don't know based on the available evidence"* for causal, off-topic or unsupported questions, and for "current air" it gives the latest sensor hour but refuses a citywide index. An optional LLM (Gemini/Groq) rewrites the verified answer in English or Urdu with [E#] citations. Code discards any output containing a number not in the evidence, a missing abstention or a fake citation.

Automation and dashboard

GitHub Actions runs every hour: unit tests (81), then the Prefect flow (warehouse + quality agents, then forecast). It commits refreshed artifacts and `prefect_runs.csv`, which Prefect state hooks write with one row per run, including failures. The Render dashboard and the Hugging Face Space show per-source freshness, quality status, the last successful run, cross-source agreement, forecast evaluation and the agent.

Limitations

PAQI's monitor set, day boundary and QA are undocumented. The Consulate monitor stopped in March 2025. Urban Unit sensors are uncorrected low-cost sensors, and their median is not a citywide or regulatory value. The forecast is a historical backtest. This is not medical advice. AI use is documented in AI_Use_Log.md.

Attribution: PAQI (Omar Abid, Dawar Hameed Butt, Mahad Naveed), CC BY 4.0 · U.S. Department of State / U.S. EPA AirNow · The Urban Unit / AirGradient, CC BY 4.0 · accessed via OpenAQ (openaq.org).